

Business Perspective

What are the business-related advantages of having the network you created?

The client-server network with Wi-Fi greatly benefits the computer shop by centralizing management of all ten PCs while also providing internet access to customers. The server allows efficient sharing of files, software updates, and system monitoring, which reduces administrative time and ensures smooth operations. Additionally, the Wi-Fi network enhances customer experience by allowing guests to connect their devices, which can attract more customers and improve the shop's reputation. The combination of wired stability and wireless convenience increases operational efficiency and overall business value.

What organizational expectations will be satisfied? Explain each.

This network fulfills several key organizational expectations. First, performance and reliability are guaranteed because the wired connections between client computers and the server provide stable and fast communication. Second, customer satisfaction is increased thanks to secure Wi-Fi access for guest devices. Third, centralized data control is achieved as all files, logs, and system settings are stored and managed through the server, preventing unauthorized access or loss. Finally, security and accountability are maintained by isolating the guest Wi-Fi from the internal LAN, ensuring that sensitive information remains protected while providing convenient public access.

Will there be any downsizing, rightsizing, upsizing opportunities that may arise in the future? If so, what are the procedures that need to be done to make that opportunity possible?

The network is designed to allow future growth and adjustments. Upsizing can be done by adding more desktop computers, gaming PCs, or additional routers for expanded Wi-Fi coverage. The process involves connecting new devices to the existing switch, updating the DHCP range for IP assignment, and configuring routers to handle additional traffic. Rightsizing could include balancing workloads between the server and client PCs to optimize performance, while downsizing might involve removing underused devices to reduce operational costs. The system's flexible design ensures that any of these changes can be implemented efficiently without major disruptions to daily operations.

Technological Perspective

What are the technology-related advantages of having the network you created?

The network combines wired and wireless technologies, providing both stability and convenience. Wired connections between the client computers and server ensure fast, reliable data transfer, while Wi-Fi allows customers to access the internet on mobile devices. The server centralizes control over software updates, file sharing, and system monitoring, which reduces management time and ensures consistent performance. Remote monitoring and administration are also possible, allowing the shop owner to oversee operations from any device connected to the network.

Are SNA, TCP/IP, or any other standards applicable to this scenario? If yes, elaborate and explain each standard.

Yes, TCP/IP is used to ensure reliable data communication between the server, client PCs, and wireless devices. Wi-Fi standards (IEEE 802.11) govern the wireless connections, providing secure, high-speed internet access for laptops and mobile devices. The Client-Server model is applied to manage requests efficiently, with client devices requesting data or services from the central server. These standards ensure interoperability, network stability, and secure management of resources across both wired and wireless connections.

Development Phases

- **Analysis:** The number of client computers and wireless devices was determined, along with their required applications, data storage needs, and internet usage patterns.
- **Design:** A star topology was chosen, with all wired devices connected to a switch and a wireless router integrated for guest internet access.
- **Development:** The server and all client computers were physically connected to the switch, IP addresses configured, and the wireless router set up to provide Wi-Fi access.
- **Testing:** Connectivity was verified between the server, wired clients, and Wi-Fi devices. Ping tests and file sharing were performed to ensure stability and performance.
- **Implementation:** The system was deployed for business operations, providing both efficient internal management and customer-facing Wi-Fi.
- **Maintenance:** Wi-Fi passwords are regularly updated, bandwidth usage is monitored, software is kept current, and security features such as firewalls and access restrictions are maintained to ensure optimal performance and safety.

Give any possible security issues that may arise at using client/server technologies in the given scenario.

Security risks include unauthorized access to the Wi-Fi network, malware from guest devices, and potential overuse of bandwidth by customers. There is also the possibility of internal misuse if staff access sensitive files improperly. These issues can be addressed by setting up a separate guest Wi-Fi network, enabling WPA2/WPA3 encryption, using firewalls, restricting internal data access, and maintaining regular software and security updates. Proper monitoring ensures both business operations and customer access remain secure.